

mentioned in the command.

```
$zenity --calendar
```

...displays the calendar dialogue box as shown in Figure 1.

The `notify-send` command allows you to send desktop notifications to the user via a notification daemon from the command line. This is useful to inform the desktop user about an event or display some form of information without getting in the user's way. To use this, you need to install the following package:

```
$ sudo apt-get install libnotify-bin
```

An example is given below:

```
$notify-send "Welcome to UNIX"
```

```
$notify-send -t 1000 -u normal "Script Running  
Successfully..."
```

where, `-t 1000` specifies the timeout in milliseconds (1000 milliseconds = 1 second)

`-u normal`: Indicates the urgency level (i.e., low, normal, or critical).

—Neethu C. Sekhar, nitucskr@gmail.com



Printing the directory structure on the terminal

Sometimes we need to print the nested directory structure of a project's source code for documentation or tutorial purposes. The tree utility comes to the rescue here. You can install it on Debian or Ubuntu using the following command:

```
$sudo apt-get install tree
```

To get the structure of the directory on the terminal, use the command given below:

```
tree directory_name
```

```
$ tree e2e
```

The output will be as shown below:

```
e2e
├─ app.e2e.ts
├─ app.po.ts
├─ tsconfig.json
└─ typings.d.ts
```

```
0 directories, 4 files
```

—Amar Shukla, amarshukla123@gmail.com



find, xargs and awk tweaks

`find`, `xargs` and `awk` are some of the powerful tools that can be used on a Linux command line. Here are a few examples of the same:

1. The following code is to archive and delete files in a folder based on a modified timestamp:

```
find /path/to/your/directory/ -maxdepth 1 -type f -mtime  
+35 -mtime -90 -print0 | xargs -0 -n 1000 bash -c 'if [ $#  
-ne 0 ]; then echo archiving $* files; tar ufp /path/to/  
archive/archive_file.tar $@ 2>/dev/null; rm -f $@; fi' bash
```

The description of this code is as follows. The `find` command will look for files with a modified timestamp older than 35 days and less than 90 days. `xargs` will process 1000 files in each batch, and the processing includes adding the files to the tar archive and removing the files from a physical location.

2. To print all the extensions in a given folder, use the following command:

```
find /path/to/your/dir/ -type f | awk -F"."  
'{a[$NF]++;print $NF}'
```

The description is as follows. To find all the files in a given directory, feed the output to `awk`, which will split the file name with "." (period) and print the last value, which is the extension.

3. To get the total size of all the files returned from `find`, use the following command:

```
find /path/to/your/dir/ -type f -exec du {} \; | awk  
'{total+=$1} END{print total}'
```

The description is as follows. Find all the files in a directory, print the size in bytes and feed the output to the `awk` command to print the total size of all files.

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